



Combining Like Terms

Day 11 of 20 · Duration: 50 minutes · Theme: Itasca State Park

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.7

Use the structure of an expression, equation and/or formula to create an equivalent form that is more helpful given the situation. Rearrange formulas to highlight a quantity of interest, using the same reasoning in solving equations. (MP6)

ESSENTIAL QUESTIONS

- What makes two terms “like”?
- Why can we add $3x$ and $5x$ but not $3x$ and 5 ?
- How do we simplify an expression by combining like terms?

VOCABULARY

Like Terms — Terms with the same variable raised to the same power. **Combine** — To add or subtract the coefficients of like terms. **Simplify** — To write an expression in its shortest equivalent form.

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can combine like terms to simplify algebraic expressions.
- I can show my work step by step and check my answers for reasonableness.

- I can explain my reasoning to a partner using math vocabulary.

ENGAGEMENT STRATEGY

Strategy: Think-Pair-Share

Practice Format: Small Group Practice

Expectation: Every student responds on a whiteboard or to a partner before any answer is shared.

BELL RINGER — WHITEBOARDS READY!

Prompt: Combine: $3x + 5x$ — what do you notice? *Students write their response on a whiteboard and hold it up. Teacher scans for misconceptions before moving on.*

LESSON PACING (50 MINUTES)

Phase 1: Bell Ringer 5 min min

Combine: $3x + 5x$ — what do you notice?

Phase 2: Direct Instruction 10 min min

Combining Like Terms — teacher-led notes (I Do); GEMDAS reference as needed.

Phase 3: Guided Practice 12 min min

Work together (We Do) — Small Group Practice. Solve together: Simplify $3x + 5 + 2x - 3$. -> Answer: $5x + 2$.

Phase 4: Culture Connection 8 min min

Liberian (Liberia) — Liberian Americans are a growing Minnesota community, many settling in Brooklyn Park and B (with math word problem)

Phase 5: Practice & Game 10 min min

GO: Simplify $3x + 5 + 2x - 3$. | SET: Simplify $7a + 4 - 2a + 6$. Simplify $4y$ squared + $3y - 2y$ squared + y . | EXTRA: Simplify $5m + 3n - 2m + 7n - 4$.

Phase 6: Exit Ticket 5 min min

Simplify: $4x + 7 - 2x + 3$.

PRACTICE PROBLEMS (GO / SET / EXTRA)

GO: Simplify $3x + 5 + 2x - 3$. **Answer:** $5x + 2$

SET: Simplify $7a + 4 - 2a + 6$. Simplify $4y$ squared $+ 3y - 2y$ squared $+ y$. **Answer:** $5a + 10$ and $2y$ squared $+ 4y$

EXTRA: Simplify $5m + 3n - 2m + 7n - 4$. **Answer:** $3m + 10n - 4$

CULTURE CONNECTION — MINNESOTA STORIES

Spotlight: Liberian (Liberia)

Liberian Americans are a growing Minnesota community, many settling in Brooklyn Park and Brooklyn Center.

Math Word Problem: A Liberian market sells rice in $4x$ -pound and $6x$ -pound bags. Simplify the total: $4x + 6x$.

Language fact: Dakota: 'Mni Sota Makoce' = 'land where the waters reflect the skies' — the origin of Minnesota's name

CAREER SPOTLIGHT — MINNESOTA

Career: CVRx — Minneapolis | Clinical Trial Coordinator | BS | \$106,184 *MN DEED 25th-percentile wage; OES-cited. Connection: algebra skills used daily on the job.*

DIFFERENTIATION

- Scaffolding: sentence starters for translation tasks; GEMDAS reference card on desks.
- Extension: students write and solve their own word problems using the lesson's operation keywords.
- Language support: bilingual vocabulary cards; pair students with a math-language peer.
- Grouping: flexible pairs/trios by readiness; Merit Time games rotate roles each round.

ASSESSMENT

- Formative: whiteboard checks during the bell ringer; circulation during GO/SET/EXTRA; exit ticket each lesson.
- Game-based: Merit Time games give instant fluency feedback (L1–L5).
- Summative: Quiz 1 (Day 7, L1–L5), Quiz 2 (Day 17, L6–L15), End-of-Unit Test (Day 20, L1–L17).